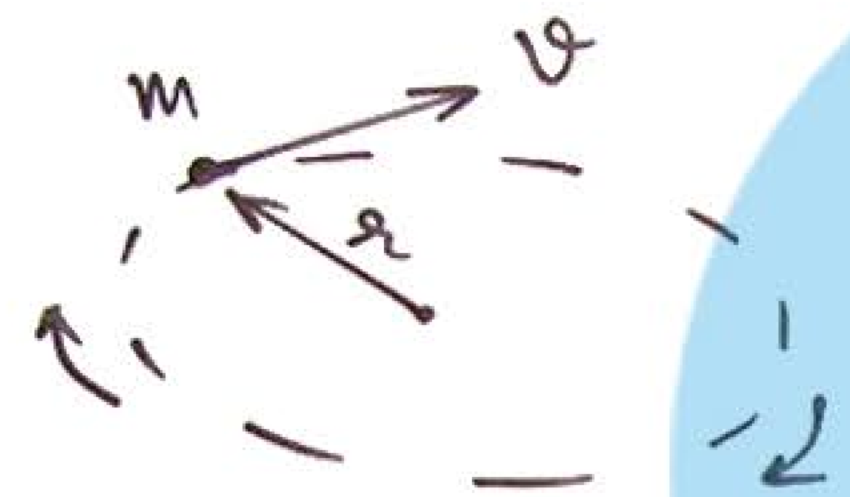
RIGID BODY DYNAMICS-PART 2

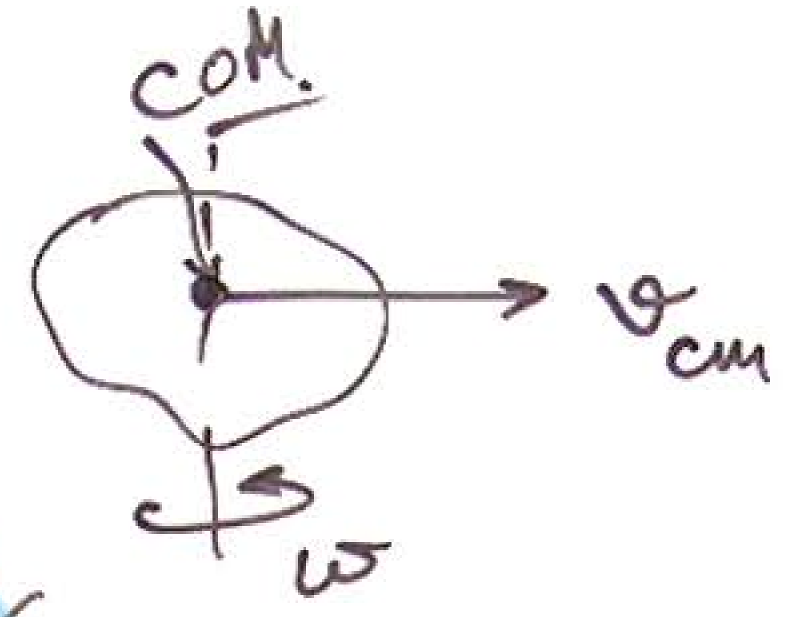
① KE in Rotational Motion :



$\omega = \frac{v}{r}$

$KE = \frac{1}{2}mv^2 \times \frac{r^2}{r^2}$

$= \frac{1}{2}(mr^2)\cdot\left(\frac{v}{r}\right)^2 = \frac{1}{2}I\omega^2$



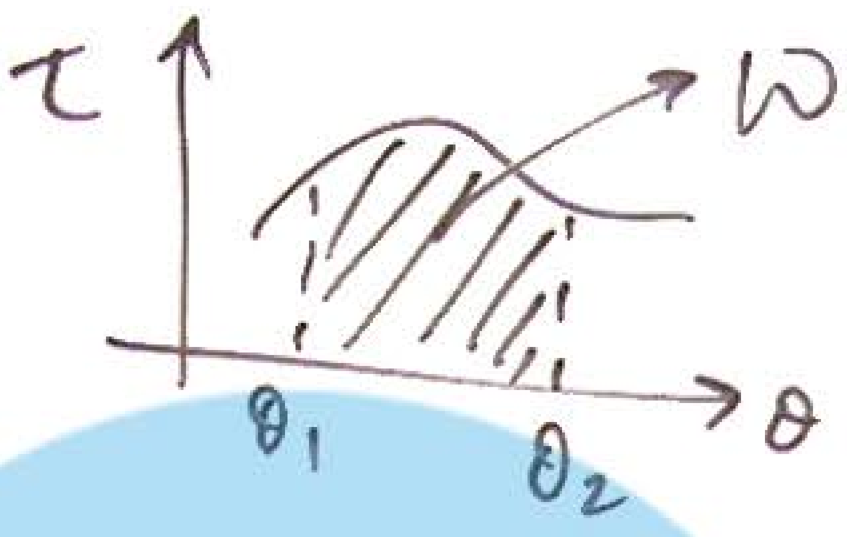
$KE = \frac{1}{2}I\omega^2 + \frac{1}{2}Mv_{cm}^2$

Valid only is AOR passes through COM.

KE of dm is $\frac{1}{2}(dm\, r^2)\omega^2 = dK$

TE of body $KE = \int dK = \frac{1}{2}\left(\int dm\, r^2\right)\omega^2 = \frac{1}{2}I\omega^2$

② Work due to Torque:

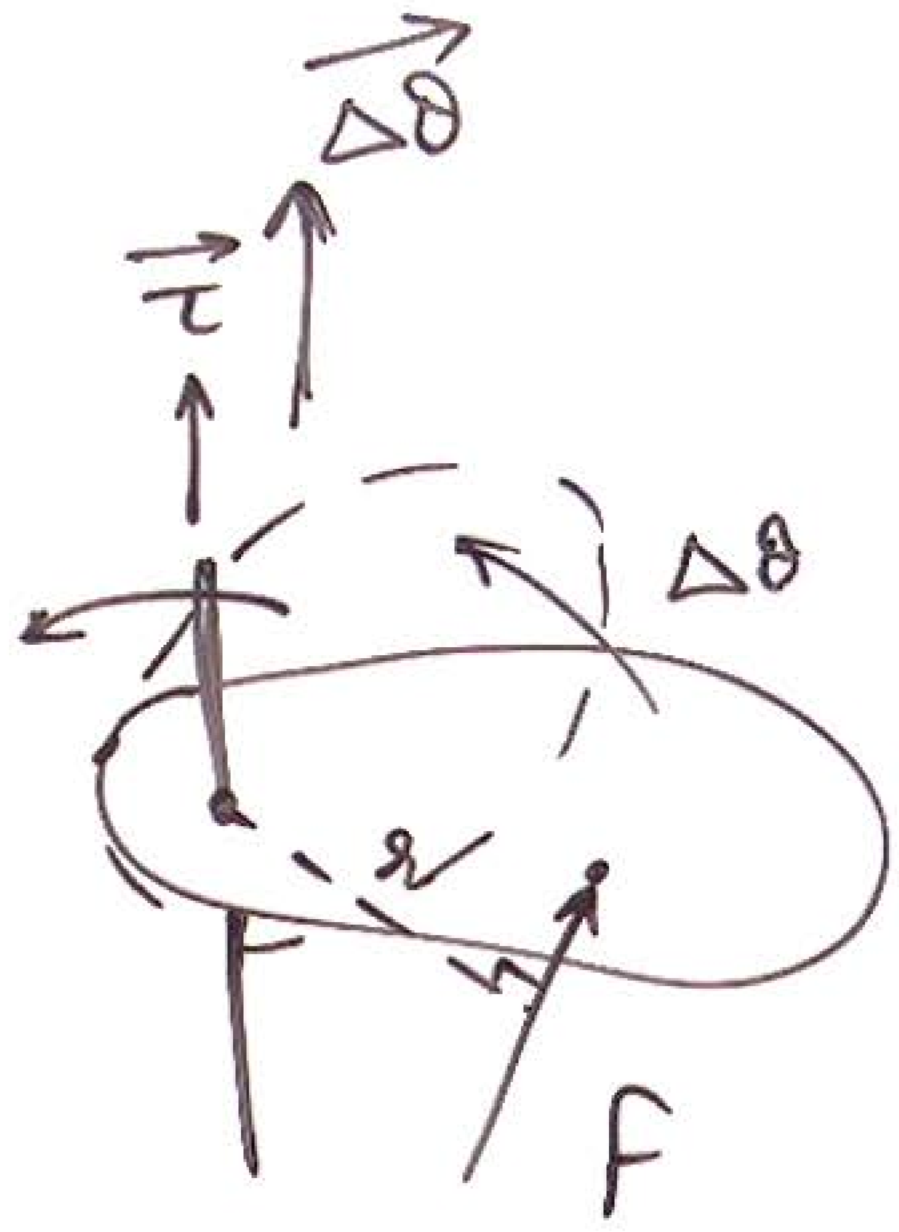


$$W = \vec{F} \cdot \vec{\Delta x} \longleftarrow \boxed{W = \vec{\tau} \cdot \vec{\Delta \theta}}$$

Angular displacement of body

for variable torque

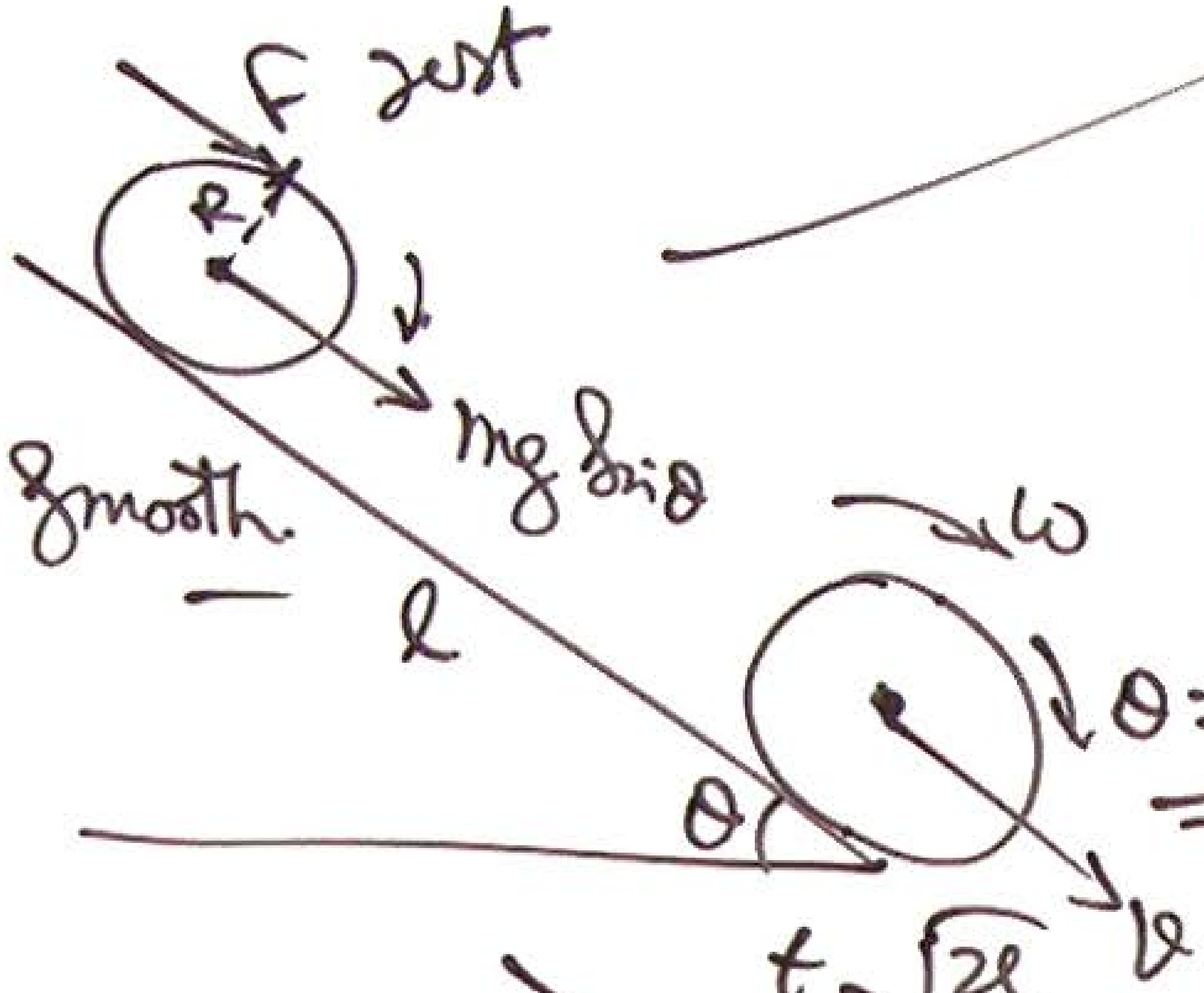
$$W = \int_{x_1}^{x_2} F \cdot dx \longleftarrow W = \int_{\theta_1}^{\theta_2} \tau \, d\theta$$



$\tau = Fr$

③ W/E Theorem in RBD:

$$\overset{K_i}{0} + \underline{mg\sin\theta \cdot l} + \underline{F \cdot l} + \underline{(F\cdot R)\cdot\theta} = \overset{K_f}{\frac{1}{2}mv^2 + \frac{1}{2}I\left(\frac{v}{R}\right)^2}$$



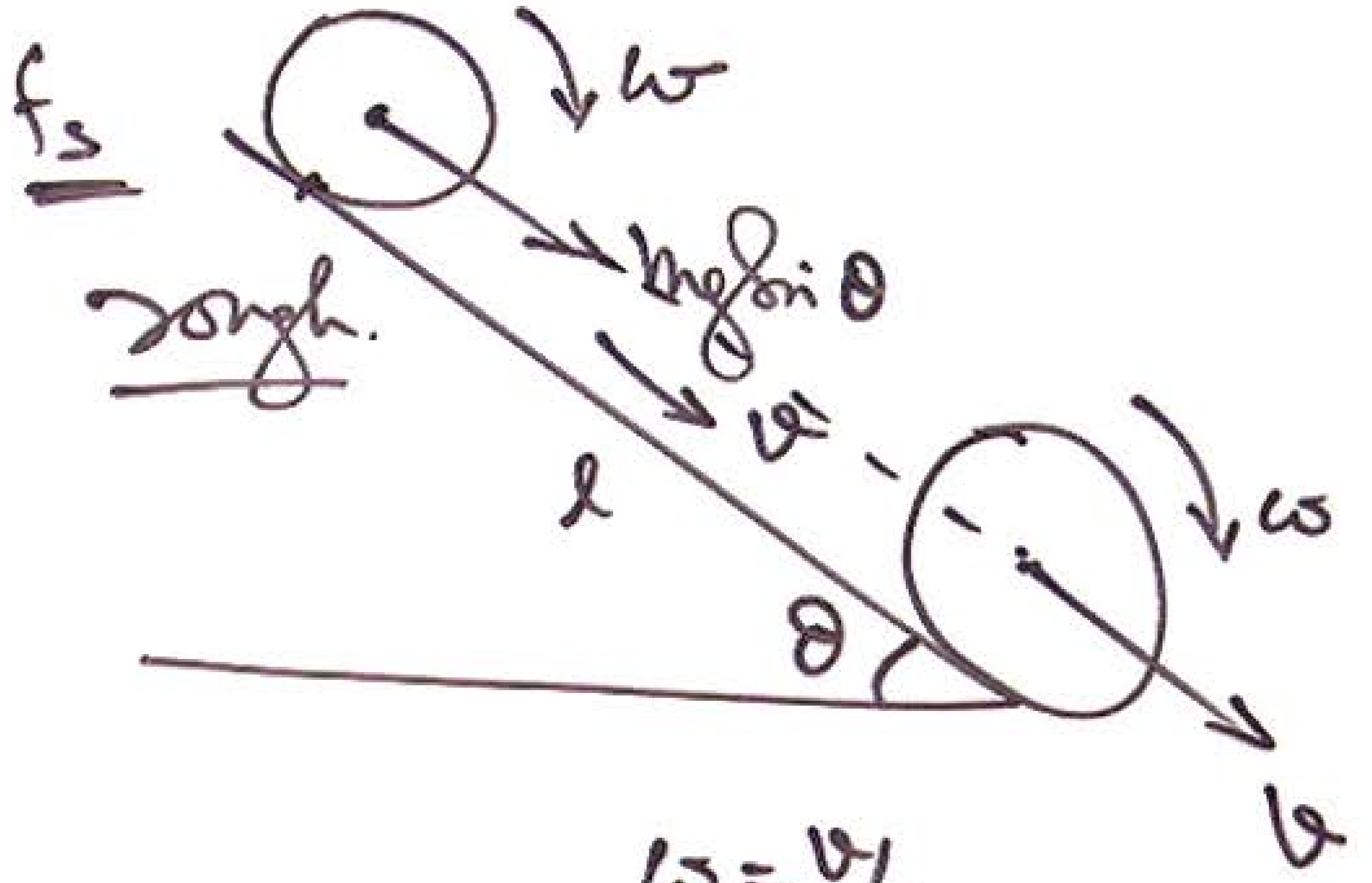
$\theta = \frac{1}{2}\alpha t^2$

$t = \sqrt{\frac{2l}{a}}$

$a = \frac{F + mg\sin\theta}{m}$

$K_i \pm$ W/D by ext forces on Body $= K_f$

W/D by a force in trans & rot^n both motions are considered separately



$\omega = v/R$

$$\overset{K_i}{0} + mg\sin\theta \cdot l = \overset{K_f}{\frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}mR^2\right)\left(\frac{v}{R}\right)^2}$$

(+R.W.D)

RIGID BODY DYNAMICS-PART 2

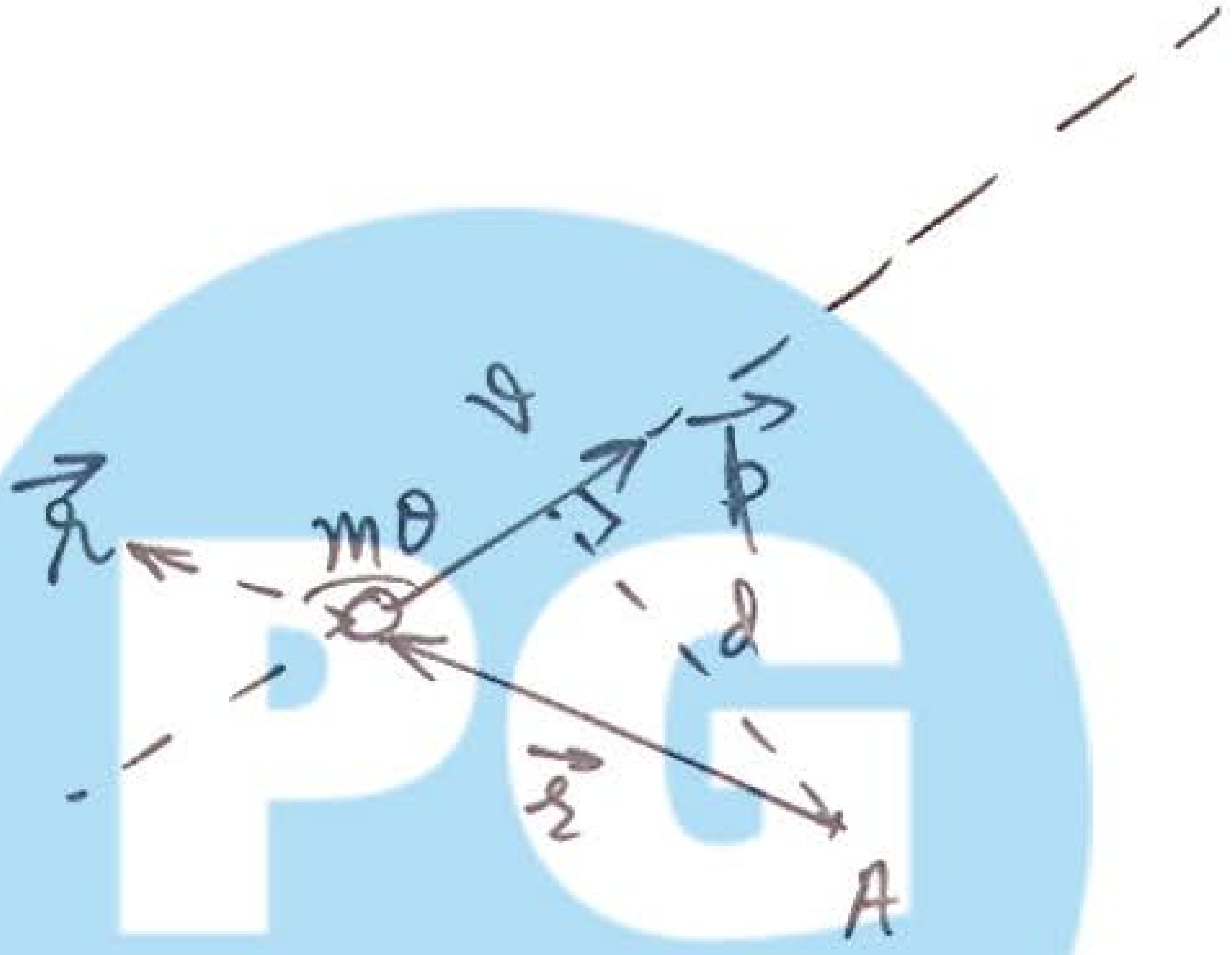
④ Angular Momentum :

Case-① linear motion.

Case-② Circular motion

Case-③ For rot^n of a rigid body ✓

$$\boxed{L = I\omega}$$



$$\left[\begin{array}{l} L_{mA} = mvd \\ \vec{L}_{mA} = \vec{r} \times \vec{p} \end{array}\right]$$

$\vec{p} = m\vec{v}$

$\vec{L}_{mA}$

$\boxed{L_{mc} = mvr} = I\omega$ about C.

$\boxed{L_{mA} = mvr_1}$ about A

Rigid Body Dynamics-Part 2

(5) Conservation of Angular Momentum :

Torque & Ang. momentum relation

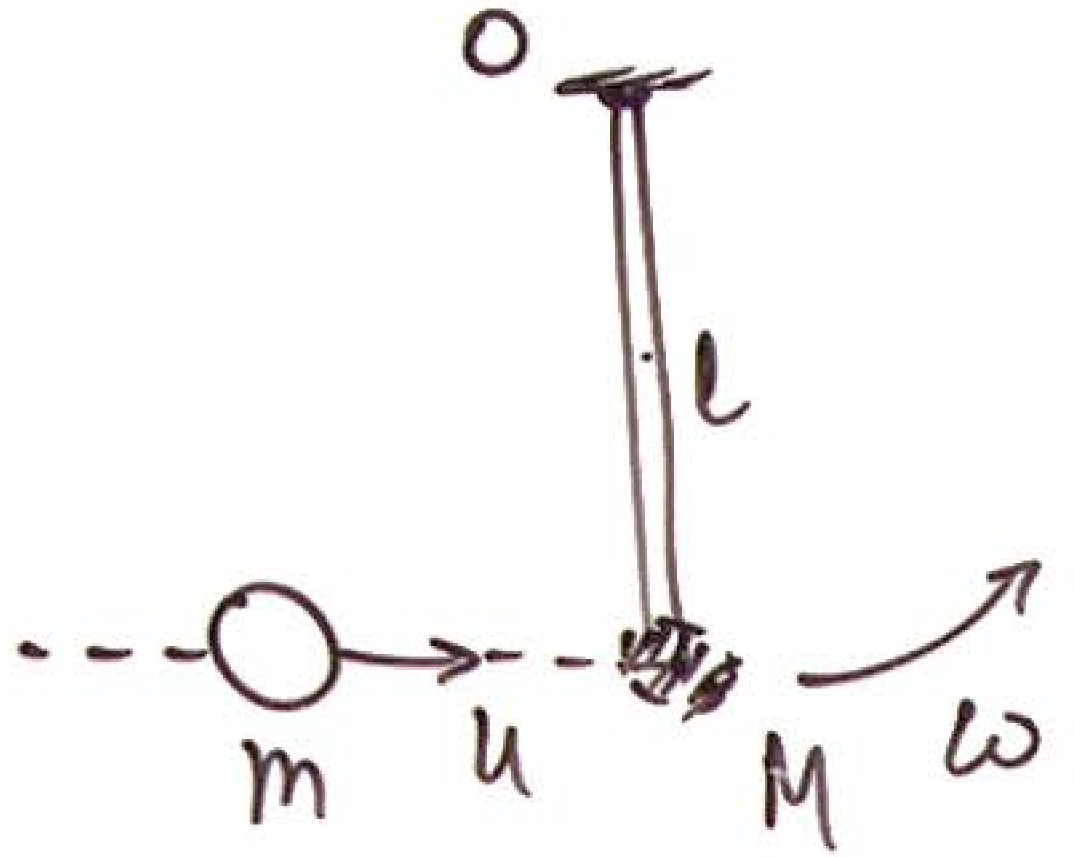
$$\boxed{\tau = \frac{dL}{dt}}$$ similar to $F = \frac{dp}{dt}$

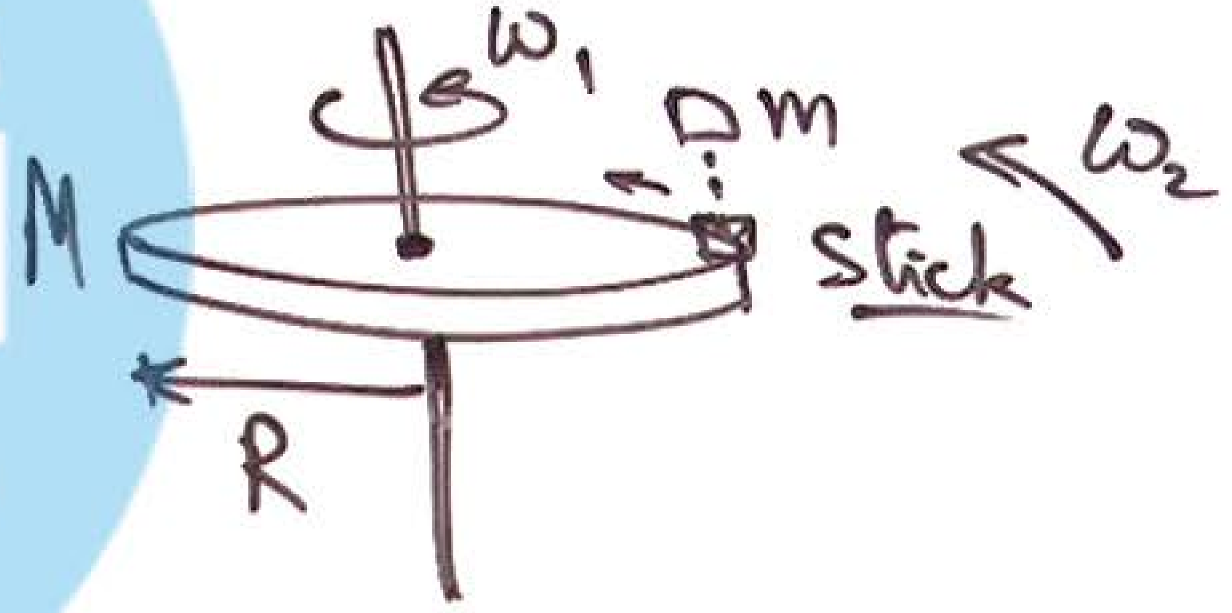
if $\tau_{ext} = 0 \Rightarrow L \rightarrow$ Conserved.

$$\left.\begin{aligned} dL &= \tau\, dt \\ \Delta L &= \int \tau\, dt \end{aligned}\right\}$$ Angular Impulse.

$$\boxed{L_i + \tau . \Delta t = L_f}$$

(6) Adding bodies to a Rotating Body :





As no ext torque is present

$\Rightarrow L_i = L_f$

$$\left(\frac{1}{2}MR^2\right)\omega_1 = \left(\frac{1}{2}MR^2 + mR^2\right)\omega_2 \Rightarrow \omega_2 = \ldots$$

By Cons of A.M. we use

$$mul = \left(\frac{Ml^2}{3} + ml^2\right)\omega$$

$$\Rightarrow \omega = \_\_\_$$

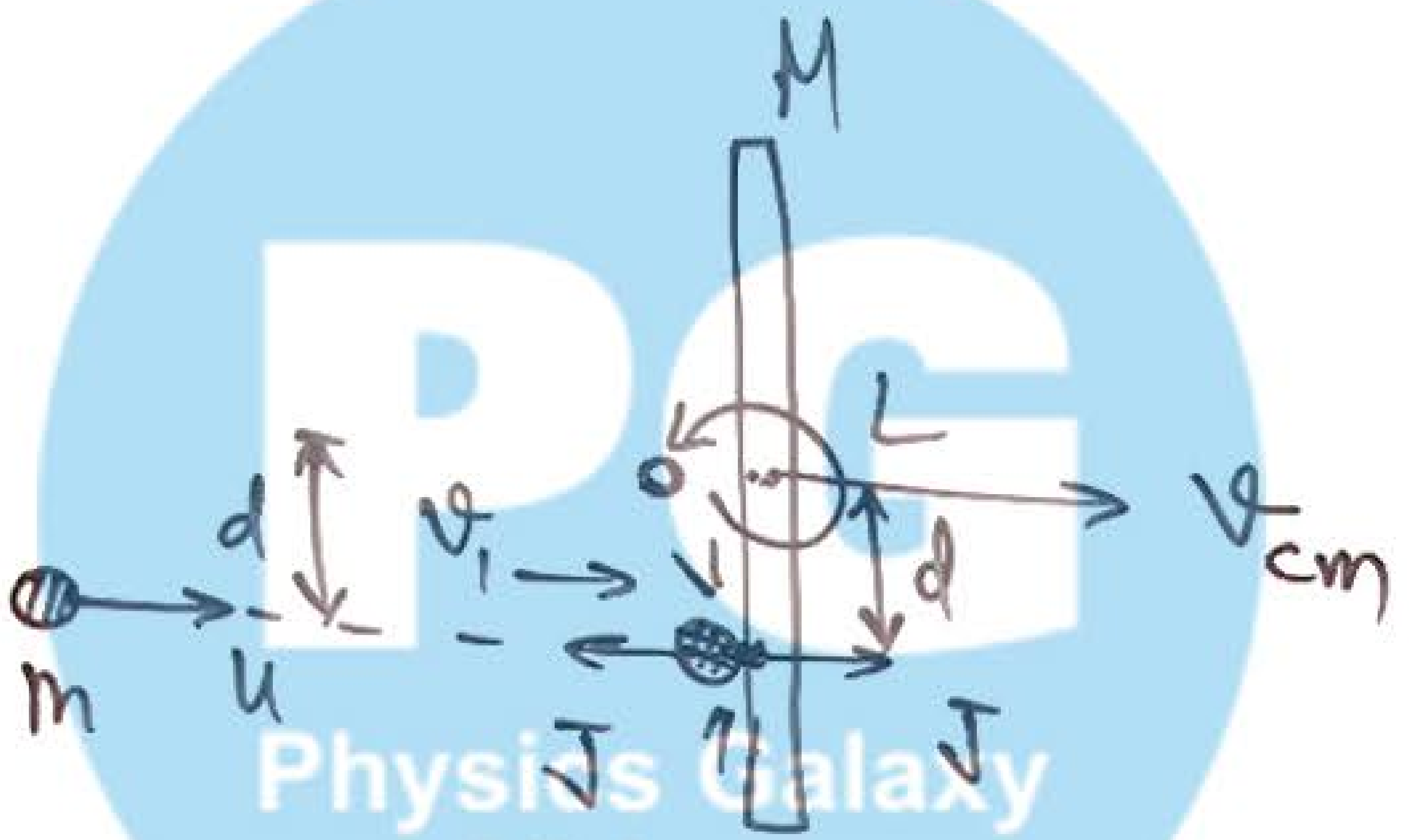
⑦ Free Rod Collisions :

By linear momentum Conservation

$$mu = mv_1 + Mv_{cm} \quad \text{—(1)}$$

By AM Conservation

$$mud = mv_1 d + \left(\frac{ML^2}{12}\right)\omega \quad \text{—(2)}$$



$e = 1$ $\quad$ $J = \int F\,dt$

by elastic Collision

$$u = (v_{cm} + d\omega) - v_1 \quad \text{—(3)}$$

Solving (1), (2) & (3)

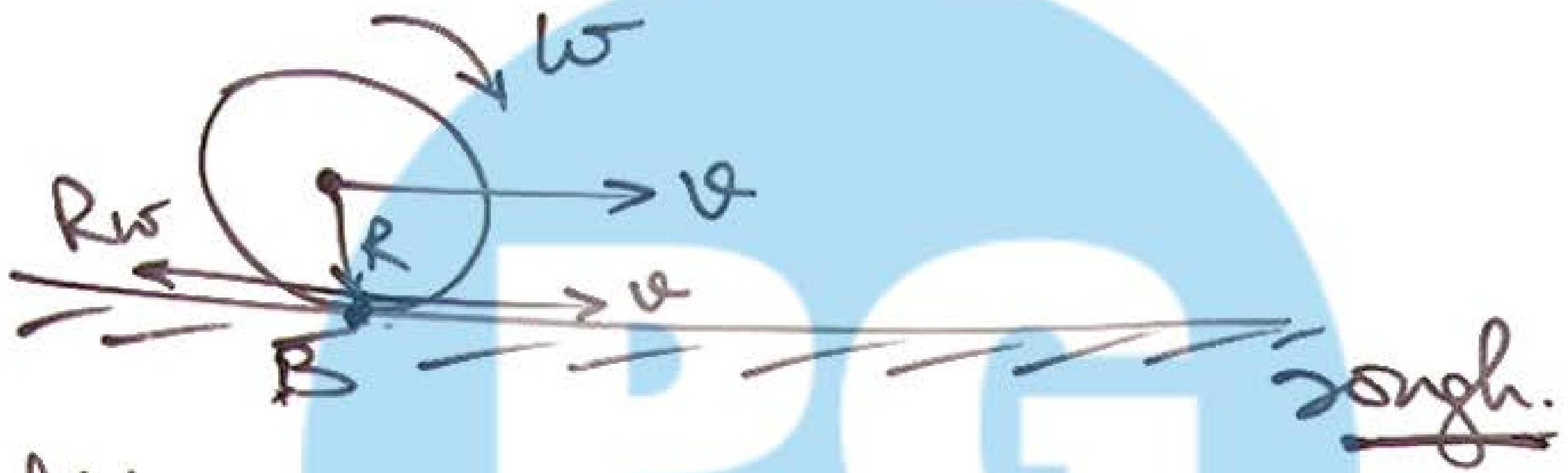
$v_1 =$ ——

$v_{cm} =$ ——

$\omega =$ ——

8 Rolling Motion:

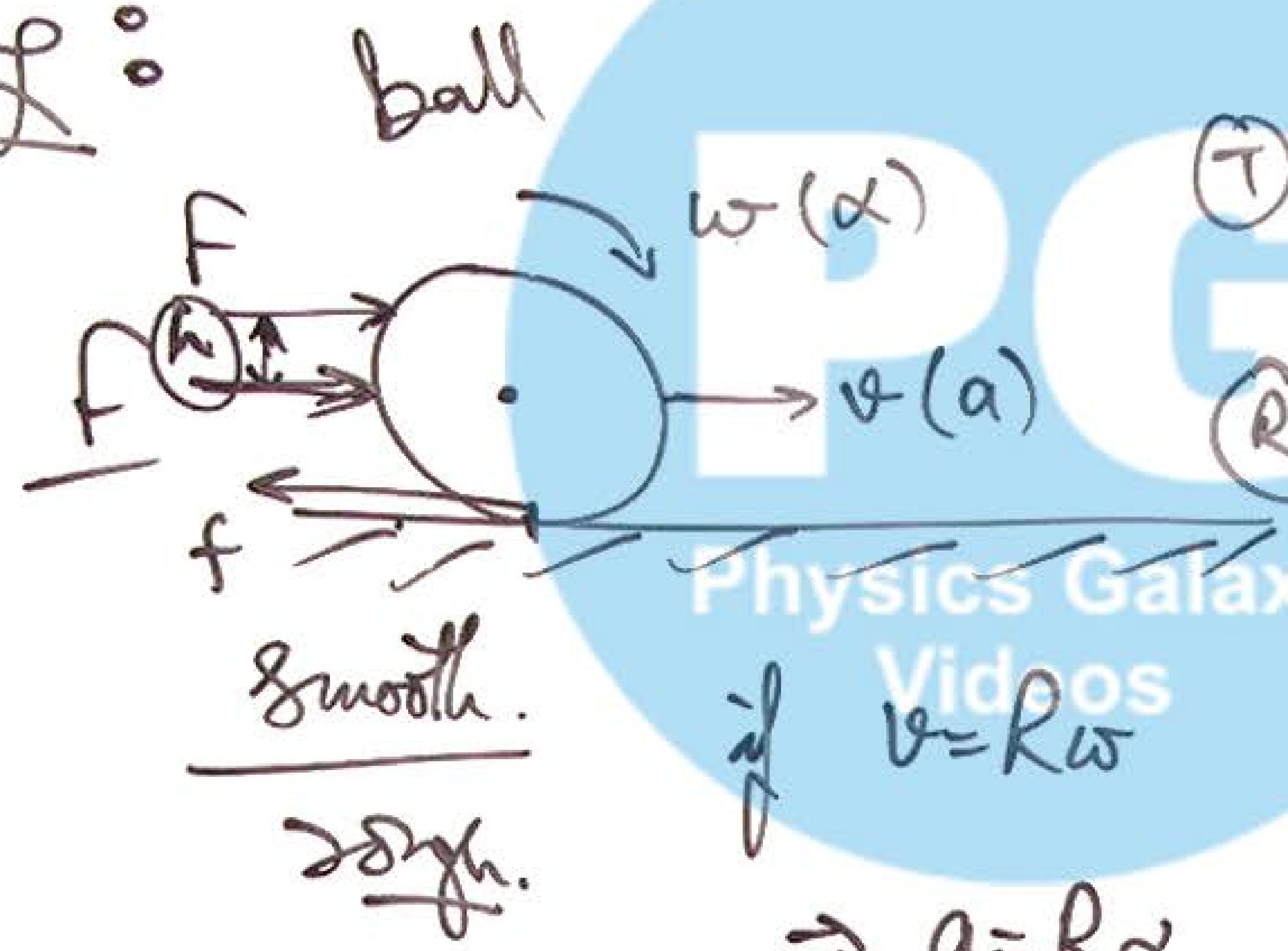
- Pure Rolling → ($v = R\omega$)



- Rolling with slipping. ($v \neq R\omega$)
  - → $v > R\omega \Rightarrow f_s$ acts backward ✓
  - → $v < R\omega \Rightarrow f_s$ acts forward. ✓

Always f_s acts in such a way so that after sometime body starts pure rolling.

9 Condition & Cases of Pure Rolling :



(T) $\rightarrow F = m R\alpha$

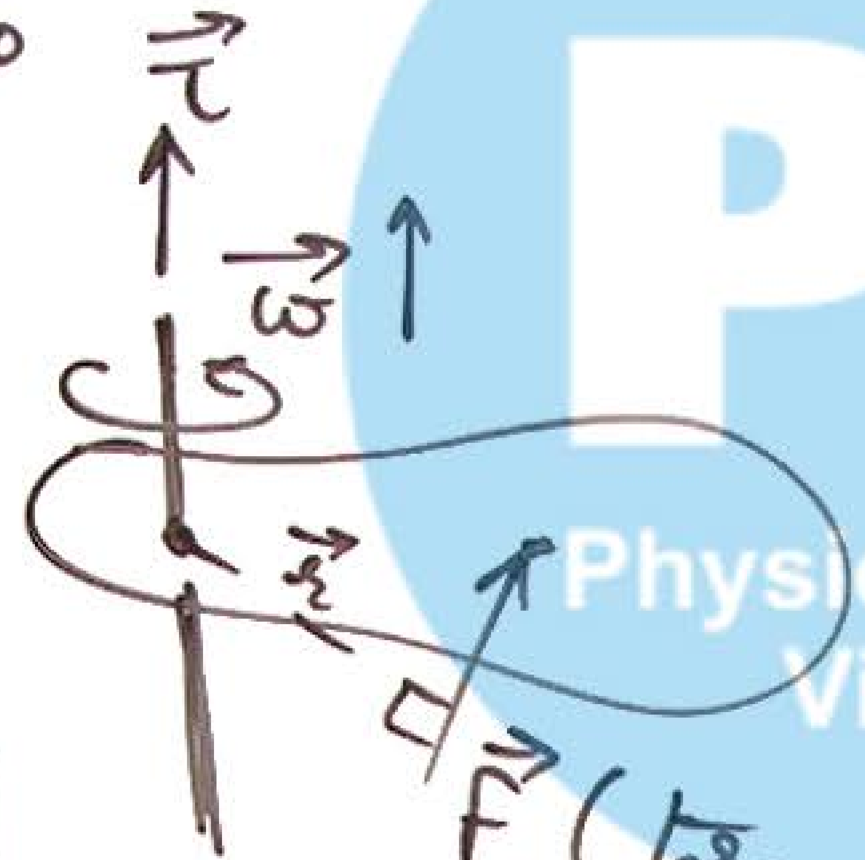
(R) $\rightarrow F \cdot h = \left(\frac{2}{5} m R^2\right)\alpha$

if $v = R\omega$

$\Rightarrow a = R\alpha$

$\Rightarrow \boxed{h = \frac{2R}{5}}$ Ans

(10) Power in Rotational Motion:



Power supplied by force

$$\boxed{P = \vec{\tau} \cdot \vec{\omega}}$$

$$\boxed{P = \tau\omega}$$ Inst Power

Power of force

$$\boxed{P = \vec{F} \cdot \vec{v}}$$

Avg power $\boxed{\langle P \rangle = \frac{\Delta K}{\Delta t}}$